Frontiers of Quantum Factoring Algorithms

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INTRODUCTION

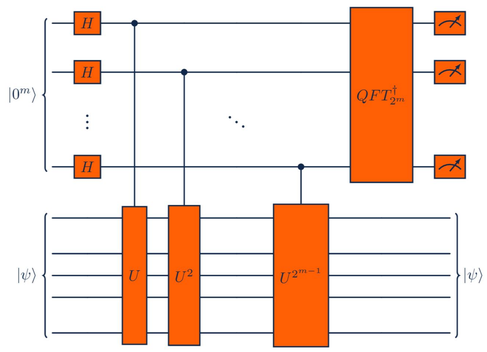
Factoring large integers is believed to be hard in classical computing. This assumed hardness is the foundation for RSA encryption. Quantum computers challenge this: Shor's algorithm (1994) factors numbers in polynomial time, and Regev's algorithm (2023) improves on Shor's asymptotically.

Quantum Fourier Transform (QFT)

- The QFT is the quantum analog of the discrete Fourier transform; it maps a state into a superposition weighted by phases.
- Requires only $O(n^2)$ gates to act on all N amplitudes at once; this is the component that extracts periodicity in both algorithms.

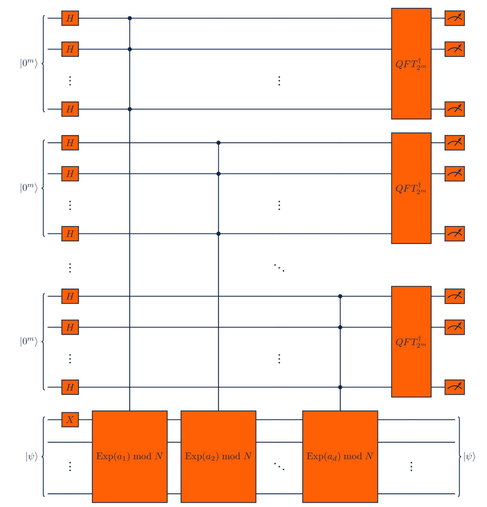
Shor's Algorithm

- Reduces factoring N to order finding: given a random number a, find the period r of $a^x \pmod N$.
- Quantum phase estimation with the inverse QFT extracts r;
- classical continued fractions recover it from the measurement.
- $\gcd(a^{r/2}\pm 1, N)$ then yields a non-trivial factor.



Regev's Algorithm

- A multidimensional generalization of Shor's algorithm
- Instead of a base a, it uses $d \approx \sqrt{n}$ small bases in superposition, so the period becomes a vector in a d-dimensional lattice.
- Fewer quantum operations per run, but requires several independent runs, plus classical LLL lattice reduction to recover the factors.
- Trades quantum cost for classical cost: shorter circuits, but the factor recovery moves into lattice reduction.



METHOD

Note: RV means Ragavan–Vaikuntanathan. We tested an RV-style filter in a small finite setting; we did not test or disprove their complete theoretical algorithm.

Shared Setup

- 8 held out semiprimes
- Fixed random seeds
- Computer simulations

The factors were hidden until the final accuracy check

Experiment 1: Regev QFT Precision

- Uniform and Gaussian Quantum Samples
- Exact QFT compared with 1 – 2 layers removed
- Samples → Lattice → LLL → GCD → Factor

Experiment 2: Shor's + Quantum Noise

- Remove small QFT rotations
- 1% Measurement Flips
- Phase and RZ errors
- 0.1% – 1% CX noise
- Fractions → GCD → Factor

Experiment 3: RV Sample Filter

- Try to identify and remove bad samples
- Test sparse and bad shots against QFT wide error
- Pass filtered samples into regev recovery
- No factors or noise labels given

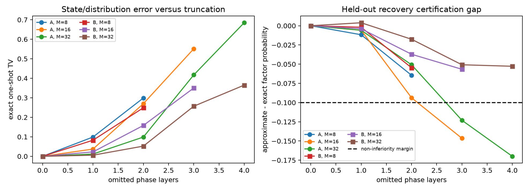
Our Work

What was measured?

- Factoring success
- Samples needed
- QFT gates
- CX gates
- Classical recovery
- Circuit depth time

RESULTS

Finite Regev Recovery Tolerates Limited QFT Truncation



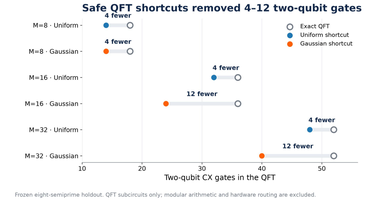
Across eight previously unseen semiprimes, removing one small QFT phase layer met our preset recovery requirement in every tested setting. Two layers passed in some Gaussian settings, while more aggressive removal reduced factoring success.

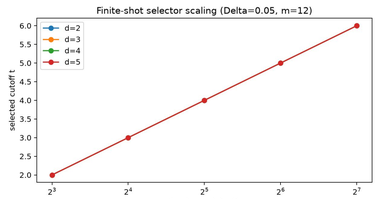
RV-Style Filtering Did Not Improve Recovery

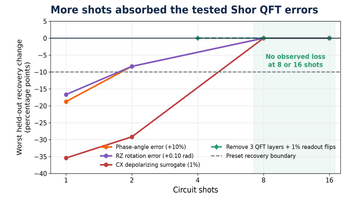
Sample Type	Standard recovery	RV-style filter
Notebook Samples	27%	15%
Gaussian Samples	22%	13%
Noisy-dual Samples	95%	95%
Corrupted Samples	25%	13%

Across 20 held-out semiprimes, our finite RV-style filter performed worse in three models and tied in one. This evaluates our implementation—not the full Ragavan–Vaikuntanathan theory.

Additional Measurements Stabilize Shor Recovery Under QFT Error







Worst of eight frozen Shor instances. Controlled simulations, including a global CX-noise surrogate; not hardware measurements.

At the largest shortcut that passed our recovery test, the QFT required 4–12 fewer CX gates. These savings apply only to the QFT subcircuit and do not establish a complete hardware speedup.

Small Circuit Walkthrough

AIM

Our aim is to compare Shor's and Regev's factoring algorithms, reduce their quantum-resource requirements, and explore whether quantum steps—especially Fourier sampling—could eventually be replaced by classical methods.

CONCLUSION & FUTURE WORK

- The choice of base a materially affects the success probability: bases with short order are cheaper to implement but more often yield trivial factors. So recourse cost of success rate trade off against each other.
- Regev shifts work from the quantum to the classical side - fewer quantum operations per run, but multiple independent runs plus LLL Basis Reduction, meaning its advantage depends on classical post-processing remaining cheap as N grows
- Results here are limited due to classical hardware; the channel ranking should be treated as indicative rather than established until tested at larger N.

We would like to expand any improvements we can make to the algorithms, by reducing the time to run, qubit size, gate size/depth. As well as looking further into dequantizing the QFT, that way it can possibly be implemented using only classical computation methods.

ACKNOWLEDGEMENTS

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REFERENCES

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Regev, Oded. "An Efficient Quantum Factoring Algorithm." *arXiv*, arXiv:2308.06572, 2023, arxiv.org/abs/2308.06572.

Other References: